Transfer: momentum and impulse

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply momentum reasoning to new collisions and deflection cases.

SUCCESS CRITERIA - I can...

1. Define the system and positive direction.
2. Use conservation or impulse only when assumptions apply.
3. Explain energy and model limits clearly.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A toy target needs a 3 km sideways shift over 5×10^{6} s. Its mass is 7×10^{8} kg. Estimate Δv and impulse; state the model used.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $p = mv$; $J = \Delta p = F_{avg}\,\Delta t$; $p_{total,\,i} = p_{total,\,f}$.
Use momentum conservation only when external impulse is negligible. Label before/after diagrams and directions.

Worked reasoning: A 2 kg cart moving at 3 m/s stops in 0.5 s. Momentum change is -6 kg·m/s; average force is -12 N. Force direction opposes the initial motion.

Your turn: Represent the same stop with momentum arrows and a force-time rectangle.

Marzano check - circle: ① ② ③

03 Represent and interpret

A 1 kg cart at 4 m/s right sticks to a 1 kg cart at rest. Draw before/after and explain which quantity is conserved.

Before contact: add velocity arrows.



Marzano check - circle: ① ② ③

04 Solve and justify

A 2 kg cart at +6 m/s sticks to a 1 kg cart at -3 m/s. Find final velocity and kinetic energy transformed.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

A report uses equal total momentum before/after to claim a collision is elastic. What additional calculation is required?

Marzano check - circle: 1 2 3

06 Choose a method

A cart has momentum -24 kg·m/s and mass 4 kg. Find velocity and its new momentum if velocity reverses at unchanged speed.

Marzano check - circle: 1 2 3

07 Exit check

A triangular force pulse has peak 40 N and duration .30 s. It acts on a 2 kg cart initially at -1 m/s. Find impulse and final velocity.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 0.4 kg ball rebounds from +5 to -3 m/s. A triangular force pulse lasts 0.08 s. Find impulse, average force and peak-force magnitude.

Marzano check - circle: 1 2 3

Read down the left column, then down the right. Use the open space to draw, calculate and explain.